INC-2642: payment unreachable, checkout orders failing

Date: 2026-06-29

Severity: P1

Duration: 16 minutes

Services affected: payment, and checkout, frontend, frontend-proxy downstream

Responders: checkout (PD-CHECKOUT)

Summary

The `paymentUnreachable` fault was active for 16 minutes. The service refuses all connections. This surfaced first on `checkout`, which is a direct caller of `payment`, and the first several minutes of the response were spent on `checkout` before the cause was localised to `payment`.

Customer impact

Checkout attempts failed at the payment step. Customers who reached the payment page saw an error and could not complete an order for the duration. Carts were preserved; no orders were lost or double-charged (payment is a stub with no real PSP).

Timeline

Time	Event
07:41	Fault injected: flag `paymentUnreachable` set to its fault variant. No alert yet.
07:43	First alert: `checkout` error ratio crosses its page threshold. checkout and the checkout team are paged.
07:45	Responder confirms the alert on `checkout` and begins checking its own dependencies rather than `checkout` itself, since `checkout` code has not changed.
07:48	Trace inspection: a `checkout` request trace shows the time originating in the `payment` span, not in `checkout`.
07:50	`payment` RED panels match the known signature — checkout sees connection-refused to payment; payment scrape target down; 100% checkout failure.
07:52	flagd checked: `paymentUnreachable` is active, set at 07:41. Cause confirmed.
07:53	Flag reset to resting variant. `payment` recovers on the next flag evaluation.
07:57	`checkout` metrics back at baseline for 3 minutes. Incident closed.

Root cause

The `paymentUnreachable` feature flag was set to its fault variant. The service refuses all connections. `payment` itself was healthy — correct version, no resource pressure, dependencies up. The failure was entirely injected at the flag layer.

Contributing factors

• The alert fires on `checkout` but the cause is one hop away. Without a dependency-aware first response, the natural move is to investigate the service that paged, which is the wrong one here.

• No alert correlates flag changes with incident start. flagd emits `flagd_flag_evaluations_total` but nothing pages on a flip, so the 07:52 flagd check was manual and late.

What went well

• Once a trace was pulled (07:48), the cause was clear within two minutes.

• The `payment` runbook's documented signature matched exactly, so no time was lost confirming.

Action items

1. Add a Grafana annotation and a low-urgency alert on `flagd_flag_evaluations_total` deltas, so a flag flip is visible on every dashboard. *(owner: platform-eng)*

2. Add "check the dependency graph before the service that paged" as step 1 of the `checkout` runbook. *(owner: checkout)* — done, see `knowledge/runbook/checkout.md`.

3. Confirm the `payment` runbook's known-failure-modes section lists `paymentUnreachable` with this signature. *(owner: checkout)*

Detection signature

```
checkout sees connection-refused to payment; payment scrape target down; 100% checkout failure.
```